

Euseok Han

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RESEARCH INTERESTS

•3D Scene Understanding, Robot Learning, and Embodied AI; ultimately, AI that understands the physical world

EDUCATION

•**Korea Advanced Institute of Science and Technology, Daejeon**

Mar. 2020 – Aug. 2027 (Expected)

B.S. in Electrical Engineering and Computer Science (Double Major)

GPA: 4.02/4.3

– *Coursework:* Machine Learning for 3D Data, Kalman Filtering (graduate), Mathematical Physics

•**University of Texas at Austin**

Aug. 2025 – Dec. 2025

Exchange Program

GPA: 4.0/4.0

PUBLICATIONS

[1] **SenseFuse: Label-Free Fusion of Image and Shape Encoders for Open-Vocabulary 3D Instance Segmentation** *Under review*

Euseok Han, Tri Ton, Hwanhee Kim, Seungyeon Ryu, Chang D. Yoo

ICRA 2027 – IEEE International Conference on Robotics and Automation [arXiv:2609.20475]

[2] **Hamiltonian Invariant-Driven Mode-Transition IMM for Satellite Maneuver Detection**

Manuscript in preparation

Euseok Han, Sangheon Choi, Seung-Hyun Kong

RESEARCH EXPERIENCE

•**Artificial Intelligence Machine Learning Lab, KAIST**

Jan. 2025 – Feb. 2025, Jan. 2026 – Present

Undergraduate Researcher / Advisor: Prof. Chang Dong Yoo

Daejeon, KR

- URP: *From Point Cloud to Actions: 3D Structural Keypoint-Conditioned Hierarchical Robot Policy Learning*
- Revealed that frozen 2D image and 3D shape encoders exhibit distinct failure patterns and weak correlation (Spearman ρ 0.08–0.23, 5.9% error overlap), in contrast to strong 2D–2D correlations (Spearman ρ 0.68, 41.4% overlap) [1]
- Devised a 2D-3D fusion framework that enhances mask labeling through sensitivity maximization
- Proposed a closed-form, label-free scene-level weight calibration method operating in 3.5 ms, achieving a median gain of +7.1 points and 93% of the oracle performance

•**NC Lab, KAIST**

Mar. 2025 – Aug. 2025

Undergraduate Researcher / Advisor: Prof. Junehwa Song

Daejeon, KR

- *Socially-appropriate Use of Noise-making Everyday Appliances in High-density Urban Housing*
- Participated in experiment design and conducted a preliminary experiment using Zigbee communication modules

•**Control and Intelligent Systems Lab, KAIST**

Jul. 2024 – Aug. 2024

Undergraduate Researcher / Advisor: Prof. Heejin Ahn

Daejeon, KR

- Reproduced Constrained Policy Optimization (ICML 2017) baselines in MuJoCo for Safe RL research.

SELECTED PROJECTS

•**LLM-Guided Autonomous Robots**

Mar. 2026 – Jun. 2026

EE40078 Introduction to Multi-disciplinary Robotics, School of Electrical Engineering

- LLM-guided autonomous mobile delivery robot; RTAB-Map localization, path planning, and object pick-up

- Hamiltonian Invariant-Driven Mode-Transition IMM for Satellite Maneuver Detection** *Mar. 2026 – Present*
MO50125 The Principles and Applications of the Kalman Filter, Graduate School of Mobility
- IMM transition probabilities driven by a measurement-side CUSUM on the J_2 -Hamiltonian Lie residual [2]

- PintOS-KAIST** *Sep. 2024 – Dec. 2024*
CS330 Operating Systems, School of Computing
- Implemented kernel modules for process scheduling, syscall, memory management of an educational x86-64 OS

HONORS AND AWARDS

- Korea-U.S. Advanced Industries Youth Exchange Scholarship** *Aug. 2025*
\$9,000 for advanced-industry exchange study in the U.S.

OTHER EXPERIENCE

- Republic of Korea Air Force (mandatory military service)** *Apr. 2022 – Jan. 2024*
Information System Operator, Staff Sergeant, KAMD Operations Center *Osan Air Base, KR*
- Operated real-time data processing servers for the Korea Air and Missile Defense (KAMD) center; 24/7 availability

TECHNICAL SKILLS AND INTERESTS

Programming Languages: C, C++, Python
Libraries: ROS, NumPy, PyTorch
Miscellaneous tools: MATLAB, L^AT_EX, Docker
Extracurricular: STROKE (Tennis Club)